

BRAC University (Department of Computer Science and Engineering)
CSE 330 (Numerical Methods)

Quiz 03

CO3

Student ID:

Section:

Name:

Full Marks: 10

Duration: 20 minutes

Solution

1. Consider the following function $f(x) = \sin(x)$ and the given node points:

x	$f(x)$	$f'(x)$
$\pi/2$	1	0
π	0	-1

- a) What will be the **degree** of the Hermite Polynomial? [2 Marks]
b) Calculate the **Hermite Bases** for the given data points. [5 marks]
c) Write the **expression** for the Hermite Polynomial and find out the value of the Hermite Polynomial at $x = 2\pi/3$. [3 Marks]

a) The degree will be 3. $P_{2n+1} = P_{2(1)+1} = P_3$

b) $P_3(x) = h_0(x)f(x_0) + h_1(x)f(x_1) + \hat{h}_0(x)f'(x_0) + \hat{h}_1(x)f'(x_1)$

$P_3(x) = h_0(x) \cdot 1 + h_1(x) \cdot 0 + \hat{h}_0(x) \cdot 0 + \hat{h}_1(x)(-1)$

$P_3(x) = h_0(x) - \hat{h}_1(x)$

$h_0(x) = [1 - 2(x - x_0)l'_0(x_0)](l_0(x))^2$

$l_0(x) = \frac{x - \pi}{\frac{\pi}{2} - \pi} = -\frac{2}{\pi}(x - \pi) = -\frac{2}{\pi}x + 2$

$l'_0(x) = -\frac{2}{\pi}$

$$h_0(x) = \left[1 - 2\left(x - \frac{\pi}{2}\right)\left(-\frac{2}{\pi}\right) \right] \left(-\frac{2}{\pi}x + 2\right)^2$$

$$h_1(x) = (x - x_1)(\ell_1(x))^2 = (x - \pi) \left(\frac{x - \frac{\pi}{2}}{\pi - \frac{\pi}{2}} \right)^2$$

$$P_3(x) = \left[1 + \frac{4}{\pi} \left(x - \frac{\pi}{2}\right) \right] \left(2 - \frac{2}{\pi}x\right)^2 - \left\{ (x - \pi) \left(\frac{2}{\pi} \left(x - \frac{\pi}{2}\right)\right)^2 \right\}$$

$$\text{At } x = \frac{2\pi}{3},$$

$$P_3(x) = \left[1 + \frac{4}{\pi} \left(\frac{2\pi}{3} - \frac{\pi}{2}\right) \right] \left(2 - \frac{4}{3}\right)^2 - \left\{ \left(-\frac{\pi}{3}\right) \left(\frac{1}{3}\right)^2 \right\}$$

$$= \frac{20}{27} - \left(-\frac{\pi}{27}\right)$$

$$= \boxed{0.8574}$$